PRIYANJLI GAUTAM

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CAREER PROFILE

AI/ML Engineer with 2.5+ years of production experience architecting Generative AI, autonomous multi-agent DAG workflows, low-latency RAG architectures, multimodal video intelligence systems, and scalable serverless MLOps. Deployed **15+ production GenAI features** across media, video intelligence, and clinical diagnostic platforms. Proven expertise in sub-second vector search indexing (CLIP/SigLIP, Qdrant, FAISS), sub-millisecond context caching (Redis Cache-Aside, PostgreSQL), and accelerated inference (16.5x faster DDIM diffusion). Published researcher in reinforcement learning (ICITEEB) and oncology biomarkers (IJCA).

PROFESSIONAL EXPERIENCE

AI/ML Engineer

Dec 2025 – Present

Matchbest Software Pvt. Ltd. *Gurgaon, India*

- Architected and deployed **Prism**, an enterprise multimodal video intelligence engine delivering in-video semantic search, scene boundary segmentation, and automated 16:9 to 9:16 vertical shorts synthesis using headless C++/FFmpeg demuxing, SigLIP visual embeddings, and Whisper, **accelerating reel turnaround by 92.8% (45 mins → 3.2 mins)** with sub-second temporal pinpointing.
- Engineered an autonomous **LangGraph-style Multi-Agent DAG Orchestrator** in FastAPI, automating multimodal intent extraction, prompt recipe generation, and parallel multi-model routing across 50+ GenAI models with live Server-Sent Events (SSE) streaming of agent reasoning traces.
- Implemented a high-throughput **Dual-Tier Context Memory Pipeline** (PostgreSQL + Redis Cache-Aside) for TRACE media services, **cutting database queries by >90%** and enabling sub-millisecond (<1ms) session reopening with full multi-turn conversational inheritance.
- Architected end-to-end **Generative Media Production Pipelines** producing full-length episodic videos (up to 60 minutes), educational micro-courses, multi-panel storyboards, and programmatic commercial ad creatives.
- Developed cloud-native **Multi-Modality Radiology AI Diagnostic Pipelines** on AWS Lambda/Docker with PACS DICOM C-STORE de-identification, TorchScript model acceleration (<30ms cold-start), and automated RSNA/FHIR R4 report generation.

Machine Learning Engineer (Intern)

Jan 2024 – Sep 2025

Slaaang Media Pvt. Ltd. *Remote / India*

- Developed and productionized personalized neural ranking and two-tower recommendation models, **boosting active user platform engagement by 25%** across high-concurrency feed streams.
- Engineered transformer-based hate-speech, toxicity, and NSFW classification systems using fine-tuned BERT/RoBERTa architectures to automate real-time community stream moderation.
- Constructed multimodal AI pipelines for automated hashtag, category, and metadata extraction from video/image feeds using vision encoders and LLM-guided context synthesis.
- Optimized inference throughput for production microservices via ONNX runtime quantization and FastAPI asynchronous request batching.

TECHNICAL COMPETENCIES

Programming	Python (3.11+), C++, SQL (PostgreSQL), Bash
Generative AI & Agents	LangGraph, Multi-Agent DAG Workflows, LLMs, RAG, Prompt Engineering, Context Engineering, Tool Calling, Structured Outputs, LoRA / QLoRA Fine-Tuning
Machine Learning & CV	PyTorch, Transformers, Computer Vision, Diffusion Models (DDPM / DDIM), CLIP / SigLIP, Whisper, CNNs, Reinforcement Learning (DQN), Recommender Systems, Scikit-learn
Vector & Data Systems	FAISS, ChromaDB, Qdrant, Redis (Cache-Aside <1ms), PostgreSQL, HNSW Vector Indexing
MLOps & Cloud	AWS (Lambda, S3, ECR), Docker, TorchScript, FastAPI, SSE / WebSockets, CI/CD (GitHub Actions), Model Deployment & Monitoring

EDUCATION

M.Tech, Computer Science & Engineering *2023 – 2025* **CGPA: 8.39**
Amity University, Noida

B.Tech, Computer Science & Engineering *2019 – 2023* **CGPA: 9.06 (Top Distinction)**
Himachal Pradesh University

Senior Secondary (CBSE) *2018* **88.0%**
Doon Valley Public School, H.P.

SELECTED ARCHITECTURE & ENGINEERING PROJECTS

Autonomous Multi-Agent DAG Orchestrator *LangGraph, Redis, FastAPI, Docker, SSE*

- Engineered an enterprise autonomous multi-agent architecture executing non-linear directed acyclic graph (DAG) pipelines with dynamic cycle detection, conditional branching, and deterministic sub-25ms inter-agent message passing.
- Integrated Redis for ephemeral state checkpointing and PostgreSQL for persistent session memory, **reducing manual human workflow orchestration overhead by 82%** with automated error self-healing.
- Implemented real-time Server-Sent Events (SSE) streaming endpoints to broadcast live agent reasoning traces and tool invocations to client applications with zero UI latency.

Accelerated Text-to-Image Generative Diffusion Pipeline *PyTorch, DDIM, Cross-Attention, CLIP, CUDA*

- Implemented a from-scratch latent diffusion architecture in PyTorch featuring a multi-scale U-Net, cross-attention conditioning, and Classifier-Free Guidance (CFG).
- Replaced traditional 1,000-step DDPM sampling with a custom 50-step deterministic Denoising Diffusion Implicit Models (DDIM) scheduler, **achieving a 16.5x speedup in generation latency (1.1s vs. 18.2s baseline)** while maintaining 14.8 FID fidelity.
- Incorporated cosine variance noise scheduling and half-precision (FP16) tensor operations, halving GPU VRAM consumption to allow high-throughput inference on consumer-grade GPUs.

Serverless High-Throughput Clinical Diagnostic Platform (RadAI) *AWS Lambda, TorchScript, Docker, DynamoDB*

- Architected a HIPAA-compliant serverless inference engine for automated preliminary X-ray and CT triage, serializing PyTorch neural vision models via TorchScript to **cut cold-start initialization to <30ms**.
- Reduced emergency radiology triage queue latency from hours down to <30 seconds for critical findings (e.g. pneumothorax), integrating automated DICOM metadata validation and FHIR clinical schema export.
- Built full CI/CD deployment pipelines using GitHub Actions and AWS ECR containerization for automated zero-downtime model updates.

Real-Time Multimodal Stream Moderation Pipeline *FastAPI, WebSockets, ViT, RoBERTa, Redis Streams*

- Engineered dual-stream real-time classification pipeline analyzing synchronized video frames and live chat transcripts simultaneously, **reducing toxic content exposure by 98.4% with <140ms end-to-end latency**.
- Designed an asynchronous sliding-window audio-visual buffer using Redis Streams to perform temporal context aggregation and eliminate false-positive spikes during sudden conversational shifts.

PEER-REVIEWED RESEARCH PUBLICATIONS

Adaptive Intelligence: Exploring Reinforcement Learning in Neural Systems *ICITEEB — Published, 2025*

- Investigated the intersection of Reinforcement Learning and Deep Learning architectures for adaptive decision-making dynamics in autonomous neural systems.
- Benchmarked Deep Q-Networks (DQN), Q-Learning, and policy gradient formulations under non-stationary environments, demonstrating enhanced convergence stability and dynamic policy adaptation.
- Indexed on EBSCO Host international scientific repository with full reproducibility benchmarks in TensorFlow and Python.

Multi-Cancer Biomarker Exploration: Leveraging Deep Learning & Survival Analysis *IJCA — Published, 2025*

- Applied deep representation learning and statistical survival modeling (Cox Proportional Hazards) to identify and validate high-confidence biomarker signatures across heterogeneous oncology genomic cohorts.
- Integrated multi-omics feature extraction and dimensionality reduction with Scikit-learn and PyTorch, achieving superior concordance index (C-index) over classical statistical baselines for patient survival risk stratification.

Learning Optimal Context Structures & Working Memory for Large Language Models *Research Manuscript — In Progress, 2026*

- Researching optimal context engineering strategies across working memory buffers, recursive hierarchical summaries, tool schema definitions, few-shot exemplars, and chain-of-thought reasoning traces.
- Benchmarking the trade-off frontier between retrieval context length, KV-cache memory pressure, token efficiency, and LLM reasoning accuracy across long-horizon agentic workflows.